

# When Does Depth Recursion Pay? A Parameter-Matched Study of Weight-Tied Hybrid Transformers on 100M Unique Words

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## Abstract

Weight-tied depth recursion trades computation for effective depth without adding learned parameters. We pre-train 21 hybrid Gated DeltaNet (GDN) + grouped-query attention (GQA) transformers from scratch on the BabyLM 2026 Strict corpus under one tokenizer, data order, and optimizer, and use them to ask *on which budget axis* recursion actually pays. Under a matched **data-exposure** budget, recursion is a robust win *on language-model loss*: two exact parameter-matched pairs improve validation loss by 0.0252 and 0.0195 nats (means over three training seeds; seed sd 0.0015 and 0.0005), and a domain-balanced re-evaluation over the full 17.4M-token dev set puts the advantage at 0.0270 and 0.0196 nats, positive in all six dev domains under all three seeds (36 of 36 cells). Grammatical accuracy does not support a claim either way: replicated over the same three seeds the mean zero-shot BLiMP gain is +0.58 (sd 2.44) and +1.84 (sd 2.05) points, against a seed-noise floor of up to 2.80 points sd for a *single* architecture retrained—so the +3.31 and +4.21 obtained from one seed are not separable from training noise. Under a matched **compute** budget it is not: the same non-recursive twins trained to 1B tokens, using two thirds of the recursive models’ training FLOPs, beat them by 0.0117 and 0.0088 nats. Two further controls sharpen the mechanism. Untying the re-applied layers—same depth, same FLOPs, three times the parameters—recovers a further 0.0467 and 0.0399 nats, pricing what tying costs. Replacing effective-depth residual scaling with unique-depth scaling changes loss by only 0.0034 and 0.0003 nats, so the gain comes from tying rather than from the initialization the two differ in. A 25M-token checkpoint grid locates the recursive models’ crossover with their twins at 75–125M tokens in the 2:1 pair, before either model has completed one pass over the corpus, ruling out repeated examples as the mechanism there;

the crossover moves later as effective depth grows, consistent with an early optimization cost. Recursion is therefore best understood as a **data-efficiency** mechanism for the fixed-corpus regime this benchmark defines, not a compute- or inference-efficiency one.

## 1 Introduction

The BabyLM challenge fixes the training data—roughly the number of words a human encounters during development—and asks what modeling choices make the most of it (BabyLM Organizing Committee, 2026). Most entries respond with data curricula or training objectives. We instead use the fixed-data setting for what it does best: making an *architectural* question unusually clean to answer.

**If parameters are scarce, is it better to spend them on independent layers, or to re-apply a smaller stack of layers several times with tied weights?**

Weight-tied depth recursion—applying the same block of layers repeatedly—decouples a model’s *effective depth* from its parameter count. The idea is old (Dehghani et al., 2019; Lan et al., 2020) and now spans from-scratch small models (Liu et al., 2024), large recurrent-depth pre-training (Geiping et al., 2025), conversion of pre-trained models (Bae et al., 2024), and adaptive recursion (Bae et al., 2025).

The question is under-specified in a way that matters. “Scarce” can mean scarce *parameters*, *compute*, or *data*, and re-applying a stack three times holds parameters fixed while tripling compute—so a parameter-matched comparison alone can only say that more layer applications beat fewer at equal weight storage, which is close to definitional. We therefore measure the same intervention against three separate budgets, and the answer differs between them.

We run the study inside a modern *hybrid* architecture that interleaves Gated DeltaNet linear-attention layers (Yang et al., 2025) with softmax grouped-query attention layers (Ainslie et al., 2023) in fixed ratios, following the design popularized by Qwen3-Next (Qwen Team, 2025). A hybrid *ratio unit* (e.g., GDN–GDN–GQA) is a self-contained token-mixing block, so recursion has an obvious granularity—repeat the unit.

## Contributions.

1. **Recursion pays per word, not per FLOP.** At matched data exposure, two exact parameter-matched pairs favour recursion by 0.0252 and 0.0195 nats. At matched compute, non-recursive twins trained longer win by 0.0117 and 0.0088 nats while using two thirds of the FLOPs. Both directions are far outside training-seed variance (§5.2, §5.3).
2. **Three seeds, six domains, and a mechanism isolated by two controls.** Every headline loss comparison is replicated across three independently seeded runs (seed sd  $\leq 0.0015$  nats) and re-measured on every dev domain rather than a 2M-token prefix; the advantage is positive in 36 of 36 domain $\times$ seed cells, and is largest on written text, smallest on speech (§5.2). An untied-depth arm prices what tying costs, and an initialization ablation shows the gain is not an artifact of effective-depth residual scaling (§5.4, §5.5).
3. **The crossover precedes the first epoch.** A 25M-token checkpoint grid places the 2:1 crossover at 75–125M tokens, before one corpus pass, eliminating repetition as an explanation; crossover time grows with effective depth (§5.6).
4. **BLiMP is too seed-noisy at this scale to carry an architectural claim.** Retraining one architecture under a different seed moves macro-accuracy by up to 5.56 points (sd 2.80), which swamps every architectural contrast in our grid; our own previously reported +3.31 does not survive replication. We report the noise floor so others can calibrate against it (§5.7).

**Evaluation note.** A training-time logger averaged per-*batch* means with batch size tied to a memory setting, biasing four logged losses by

about  $-0.025$ . That logger has since been fixed to weight by rows; every validation number below is recomputed from checkpoints with uniform per-token weighting regardless. Appendix A documents the pitfall, its fix, and the calibration confirming both.

## 2 Related Work

**Parameter sharing across depth.** Universal Transformers (Dehghani et al., 2019) apply one block recurrently with adaptive halting; ALBERT (Lan et al., 2020) ties encoder layers. MobileLLM-LS pre-trains 125M- and 350M-parameter decoder-only models with immediate block-wise sharing (Liu et al., 2024). Geiping et al. (2025) scale a recurrent-depth decoder to 3.5B parameters. Relaxed Recursive Transformers convert pre-trained models and soften ties with layer-wise LoRA (Bae et al., 2024), while Mixture-of-Recursions learns token-specific depths (Bae et al., 2025). These studies establish that sharing works; none separates the parameter, compute, and data axes under a fixed corpus, which is what the BabyLM setting makes possible.

**Subquadratic token mixers at BabyLM scale.** This is not the first BabyLM entry to replace or dilute softmax attention. Haller et al. (2024) introduced BabyHGRN, which uses the HGRN2 recurrent architecture and outperformed transformer and other subquadratic baselines (LSTM, xLSTM, Mamba) under the 2024 Challenge budgets. Their follow-up BLaLM (Haller et al., 2025) replaces self-attention with a linear-time mLSTM token mixer plus short convolutions, sliding-window attention, and Hedgehog feature maps, reporting that linear attention combined with sliding-window attention improves zero-shot performance and that Muon converges more stably than AdamW. Our setting differs in two ways that matter: we retain a *minority* of full softmax layers rather than eliminating attention, and our object of study is the re-application schedule of a fixed layer inventory rather than the mixer itself. Their results are the relevant prior for the viability of subquadratic mixers at 100M words; ours addresses a question orthogonal to mixer choice.

**Linear attention and the delta rule.** DeltaNet-style linear attention maintains a fast-weight state updated by the delta rule (Yang et al., 2024). Gated DeltaNet adds a data-dependent decay gate

(Yang et al., 2025). We use the reference Triton kernels from flash-linear-attention (Yang and Zhang, 2024).

**Hybrid architectures.** A minority of softmax layers interleaved with a majority of linear-attention layers preserves most of full attention’s quality at a fraction of its inference cost; Qwen3-Next (Qwen Team, 2025) adopts a 3:1 ratio at 80B scale. We evaluate 2:1 and 3:1 at BabyLM scale and use the ratio unit as the unit of recursion.

**Sample-efficient pre-training.** The BabyLM series (Warstadt et al., 2023; BabyLM Organizing Committee, 2026) fixes the corpus (Strict: 100M words; repetition allowed up to 1B words of exposure). Our primary runs consume  $\sim 295$ M words each; the compute-matched arms consume  $\sim 589$ M and  $\sim 884$ M, all within budget.

### 3 Models

All variants share one spine: a decoder-only pre-norm transformer with  $d_{\text{model}} = 768$ , RMSNorm (Zhang and Sennrich, 2019,  $\epsilon = 10^{-5}$ ), SwiGLU feed-forward blocks (Shazeer, 2020) with  $d_{\text{ff}} = 2304$  (a fixed  $3\times$  multiplier for *all* variants), tied input/output embeddings, and the BabyLM-community BPE tokenizer (vocabulary 16,384).

**GQA layers** use 12 query heads and 4 KV heads of dimension 64, rotary position embeddings (Su et al., 2021) with base 10,000, and per-head RMS QK-normalization (Henry et al., 2020). Softmax attention runs FlashAttention-3 (Shah et al., 2024) on Hopper with an SDPA fallback.

**GDN layers** are Gated DeltaNet layers (Yang et al., 2025) as implemented in flash-linear-attention v0.5.1: 12 heads of dimension 64, value expansion 1, output gating and short (size-4) depthwise convolutions enabled.

#### 3.1 The variants

A *ratio unit* is (GDN, GDN, GQA) for 2:1 and (GDN, GDN, GDN, GQA) for 3:1. The non-recursive hybrids stack two units. The recursive hybrids contain **two independent super-blocks in sequence, each holding one ratio unit, each applied  $R = 3$  times with weights tied within that super-block** (never across super-blocks). Unrolled, Recursive 2:1 computes 18 layer applications with the parameters of 6 layers.

| Model         | s42   | s43   | s44   | sd   | range       |
|---------------|-------|-------|-------|------|-------------|
| GDN 2:1       | 62.24 | 65.60 | 67.80 | 2.80 | <b>5.56</b> |
| Recursive 2:1 | 65.56 | 64.22 | 67.62 | 1.71 | 3.39        |
| GDN 3:1       | 63.27 | 66.05 | 65.19 | 1.42 | 2.78        |
| Recursive 3:1 | 67.48 | 66.77 | 65.78 | 0.85 | 1.70        |

Table 1: BLiMP macro-accuracy of one architecture across three training seeds. The same models’ validation loss varies by  $\leq 0.0020$  nats.

| Dev domain      | Tokens | Wins. | 2:1            | 3:1            |
|-----------------|--------|-------|----------------|----------------|
| BNC Spoken      | 1.74M  | 424   | +0.0267        | +0.0207        |
| CHILDES         | 5.84M  | 1426  | +0.0139        | +0.0099        |
| Gutenberg       | 3.88M  | 946   | <b>+0.0406</b> | <b>+0.0310</b> |
| OpenSubtitles   | 3.39M  | 828   | +0.0238        | +0.0190        |
| Simple Wiki     | 2.29M  | 559   | <b>+0.0405</b> | +0.0267        |
| Switchboard     | 0.25M  | 60    | +0.0164        | +0.0100        |
| Domain-balanced | 17.39M | 4243  | <b>+0.0270</b> | <b>+0.0196</b> |

Table 2: Paired recursion advantage in nats by dev domain, mean over three seeds (seed sd  $\leq 0.0048$  throughout). Positive favours the recursive model in every cell—36 of 36 domain $\times$ seed $\times$ family combinations.

Table 3 lists the inventory. Within each matched pair the two models have identical learned layer inventories and parameter counts *to the digit*; src/common/param\_count.py is the executable source of truth and asserts this, along with the untied arms’  $3\times$  layer scaling and the ablations’ parameter identity.

#### 3.2 Initialization, and what it does not explain

Weights are drawn from  $\mathcal{N}(0, 0.02^2)$ ; residual-output projections are scaled by  $1/\sqrt{2D_{\text{eff}}}$ , where  $D_{\text{eff}}$  is the *effective* depth—18 or 24 for the recursive models, the plain layer count otherwise. Using effective rather than unique depth is a principled stability choice because the same projection writes into the residual stream  $R$  times.

The matched pairs therefore differ at initialization as well as in execution, so the comparison as originally constructed identifies the recursive *configuration* rather than tying alone. The *unique init* arms separate the two: parameter-identical to the recursive models to the digit, differing only in scaling residual-output projections by  $1/\sqrt{2 \cdot \text{unique depth}}$ , a  $\sqrt{3}$  difference. Initialization accounts for 0.0034 and 0.0003 nats against recursion gains of 0.0252 and 0.0195 (§5.5)—real but small.

The untied deep arms use  $1/\sqrt{2n_{\text{layers}}}$  with  $n_{\text{layers}} = 18$  or 24, which *equals* their recur-

| Variant                    | Mix      | Uniq. layers | Eff. depth | Non-emb. params   | Role                       |
|----------------------------|----------|--------------|------------|-------------------|----------------------------|
| GQA baseline               | pure GQA | 10           | 10         | 68,830,208        | no-GDN reference           |
| GDN 2:1                    | 2:1      | 6            | 6          | 46,913,888        | twin of Recursive 2:1      |
| GDN 3:1                    | 3:1      | 8            | 8          | 63,487,504        | twin of Recursive 3:1      |
| <b>Recursive 2:1</b>       | 2:1      | 6            | 18         | <b>46,913,888</b> | 2 SB $\times$ $R=3$        |
| <b>Recursive 3:1</b>       | 3:1      | 8            | 24         | <b>63,487,504</b> | 2 SB $\times$ $R=3$        |
| Untied 2:1 deep            | 2:1      | 18           | 18         | 140,740,128       | depth/FLOP-matched, untied |
| Untied 3:1 deep            | 3:1      | 24           | 24         | 190,460,976       | depth/FLOP-matched, untied |
| Recursive 2:1 (uniq. init) | 2:1      | 6            | 18         | 46,913,888        | initialization ablation    |
| Recursive 3:1 (uniq. init) | 3:1      | 8            | 24         | 63,487,504        | initialization ablation    |

Table 3: Model inventory. GDN 2:1 / Recursive 2:1 and GDN 3:1 / Recursive 3:1 are exact parameter-matched pairs. The untied arms deliberately are not parameter-matched: they hold depth and FLOPs fixed instead. Compute-matched arms are not new models—they are GDN 2:1 and GDN 3:1 trained to larger token budgets (§4).

sive counterparts’  $1/\sqrt{2D_{\text{eff}}}$ . Untied and recursive arms therefore share an initialization scale by construction and differ only in whether weights are tied.

## 4 Experimental Setup

**Data.** We train on the official BabyLM 2026 Strict corpus (six domain files,  $\sim 100\text{M}$  words) and validate on the official dev split, excluded from the training bins at preprocessing time. Under the shared tokenizer one epoch is 169,741,563 tokens ( $\approx 1.70$  tokens per whitespace word).

**Recipe.** One locked recipe for every run: 4,096-token sequences, 32 sequences per optimizer step, AdamW at peak  $6 \times 10^{-4}$  with cosine decay to 10%, and a 499,908,608-token primary budget ( $\approx 2.9$  epochs  $\approx 295\text{M}$  words). Appendix A gives the full hyperparameters and hardware.

**Seeds.** Each matched pair is trained at seeds 42, 43 and 44. A seed sets weight initialization *and* data order together, so the control holds within a seed group while the spread across groups is genuine run-to-run training variance.

**Compute-matched arms.** A recursive model executes  $\sim 3\times$  the layer FLOPs per token of its twin, so at 500M tokens it consumes  $\approx 1.5\text{B}$  token-equivalents of training compute. We therefore train GDN 2:1 and GDN 3:1 to **1.5B tokens** (11,444 steps, 750 warmup,  $\approx 884\text{M}$  words—exact FLOP parity) and to **1B tokens** (7,629 steps, 500 warmup,  $\approx 589\text{M}$  words—two thirds of parity), each with the cosine schedule stretched to its own horizon and the warmup *fraction* held at  $\approx 6.5\%$ . Both remain inside the Strict track’s 1B-word exposure cap. The 1B arm exists because the 1.5B arm overtrains (§5.3), so reporting only the FLOP-parity endpoint would flatter recursion.

**Evaluation protocol.** All validation numbers are recomputed from retained checkpoints over the same 488 non-overlapping 4,096-token windows, with uniform per-token weighting and an identical code path, for all 21 runs and 255 checkpoints. The 2M-token prefix was fixed to bound cost, not selected by model performance; because sorted-file concatenation makes it BNC Spoken plus part of CHILDES, final checkpoints are additionally scored on all six dev domains (§5.2).

## 5 Results

### 5.1 Overall picture

Read down Table 4’s loss column and the ordering is not by parameters, not by effective depth, and not by tokens—it is by how the budget was spent. The rest of this section separates those axes.

### 5.2 Recursion at matched exposure

At the shared 500M-token budget the recursive member wins both pairs, and the effect replicates (Table 5). Within seed 42, a paired 100,000-resample bootstrap over the 488 aligned windows gives 0.0261 [0.0248, 0.0274] and 0.0195 [0.0183, 0.0207] nats, with recursion winning 480/488 and 462/488 windows (exact sign tests  $p = 1.92 \times 10^{-130}$  and  $2.64 \times 10^{-104}$ ). The published single-seed values sit essentially on the three-seed means. Establishing the seed floor at  $\leq 0.002$  nats is what makes the smaller effects below interpretable rather than suggestive.

**The advantage holds in every dev domain.** Because preprocessing concatenates the six dev files in sorted-filename order, the fixed 2M-token prefix used above is 87% BNC Spoken and 13% CHILDES; four of six domains contribute nothing to it. We therefore re-scored the final check-

| Model                      | Non-emb. params | Tokens | Val loss ↓    | Val ppl ↓    | bpt ↓        | BLiMP ↑      | Suppl. ↑     |
|----------------------------|-----------------|--------|---------------|--------------|--------------|--------------|--------------|
| Untied 3:1 deep            | 190.5M          | 500M   | <b>3.0527</b> | <b>21.17</b> | <b>4.404</b> | 67.50        | 59.53        |
| Untied 2:1 deep            | 140.7M          | 500M   | 3.0605        | 21.34        | 4.415        | 68.17        | 59.26        |
| GDN 3:1 (1B, annealed)     | 63.5M           | 1B     | 3.0838        | 21.84        | 4.449        | 67.71        | 55.95        |
| <b>Recursive 3:1</b>       | 63.5M           | 500M   | 3.0926        | 22.03        | 4.462        | 67.48        | 55.17        |
| Recursive 3:1 (uniq. init) | 63.5M           | 500M   | 3.0929        | 22.04        | 4.462        | 67.54        | 54.55        |
| GDN 2:1 (1B, annealed)     | 46.9M           | 1B     | 3.0955        | 22.10        | 4.466        | 65.58        | 57.95        |
| <b>Recursive 2:1</b>       | 46.9M           | 500M   | 3.1072        | 22.36        | 4.483        | 65.56        | 53.30        |
| GDN 2:1 (1.5B)             | 46.9M           | 1.5B   | 3.1081        | 22.38        | 4.484        | 67.56        | <b>60.68</b> |
| Recursive 2:1 (uniq. init) | 46.9M           | 500M   | 3.1106        | 22.43        | 4.488        | 65.75        | 56.08        |
| GDN 3:1 (1.5B)             | 63.5M           | 1.5B   | 3.1120        | 22.47        | 4.490        | 66.82        | 56.32        |
| GDN 3:1                    | 63.5M           | 500M   | 3.1121        | 22.47        | 4.490        | 63.27        | 53.93        |
| GQA baseline               | 68.8M           | 500M   | 3.1244        | 22.75        | 4.508        | <b>71.07</b> | 59.59        |
| GDN 2:1                    | 46.9M           | 500M   | 3.1333        | 22.95        | 4.520        | 62.24        | 52.28        |

Table 4: All arms on the fixed dev slice, uniform per-token weighting, seed 42. Every model trained on the same 100M-word Strict corpus; the Tokens column is exposure, not corpus size. The GQA baseline used the identical 500M-token budget as the other primary runs—only the arms marked 1B and 1.5B consumed more.

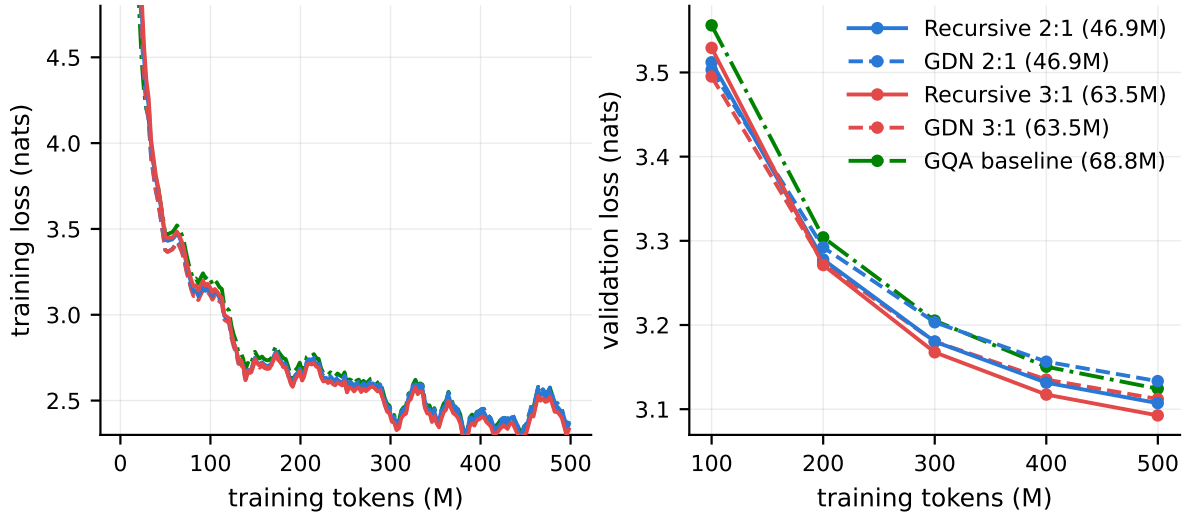


Figure 1: Training loss (left, smoothed, from training logs) and checkpoint validation loss (right, uniform protocol) versus training tokens for the five primary variants. Hue encodes the ablation pair (blue = 2:1, red = 3:1, green = baseline); solid lines are recursive models, dashed their non-recursive twins. All 21 runs in the study are stable, with zero loss spikes.

| Pair | s42     | s43     | s44     | mean           | sd     |
|------|---------|---------|---------|----------------|--------|
| 2:1  | +0.0261 | +0.0236 | +0.0261 | <b>+0.0252</b> | 0.0015 |
| 3:1  | +0.0195 | +0.0200 | +0.0190 | <b>+0.0195</b> | 0.0005 |

Table 5: Paired recursion advantage (GDN minus Recursive) in nats, replicated over three independently seeded training runs. Per-model seed sd ranges from 0.0005 to 0.0020.

point of all thirteen 500M-token runs on the entire 17,390,678-token dev set, in windows contained within a single domain. The recursive model wins in **36 of 36** domain×seed×family cells; the domain-balanced advantage is +0.0270 (2:1) and +0.0196 (3:1), with per-domain val-

ues in Table 2 (Appendix A). Two things follow. First, the narrow slice was *conservative, not flattering*: the domain-balanced advantage (+0.0270, +0.0196) slightly exceeds the prefix figure (+0.0252, +0.0195), because the prefix is dominated by the two domains where recursion helps least. Second, the effect is domain-structured in the way a depth account predicts—roughly 3× larger on edited written text (Gutenberg, Simple Wiki) than on conversational speech (CHILDES, Switchboard).

On grammatical minimal pairs the single-seed picture appeared to agree: +3.31 [0.38, 6.85] points for the 2:1 pair and +4.21 [2.43, 6.11] for the 3:1, with the 3:1 effect directionally broad (46

| Family | Recursive<br>(500M, 3 $\times$ ) | Twin 1.5B<br>(parity) | Twin 1B<br>( $\frac{2}{3}$ parity) |
|--------|----------------------------------|-----------------------|------------------------------------|
| 2:1    | 3.1072                           | 3.1081                | <b>3.0955</b>                      |
| 3:1    | 3.0926                           | 3.1120                | <b>3.0838</b>                      |

Table 6: Validation loss under matched training FLOPs. The 1B-token arm uses two thirds of the recursive models’ compute and still wins both families.

paradigms won, 20 lost, one tied;  $p = .0019$ ) and the 2:1 mean driven by a positive tail (37 won, 30 lost,  $p = .464$ ). Replicating both pairs across all three seeds dissolves most of this; §5.7 reports what is left.

### 5.3 Recursion at matched compute: the advantage reverses

Inverting the parameter-matched comparison, we hold compute fixed and let exposure vary (Table 6). At exact FLOP parity the 2:1 comparison is a tie (0.0009 nats) and the 3:1 favours recursion (0.0194). But that endpoint is misleading in the twin’s disfavour: both 1.5B runs pass their minimum near 1.0B tokens (2:1 best 3.0922; 3:1 best 3.0733) and degrade thereafter as repetition accumulates past  $\sim 6$  epochs. Trained to 1B tokens with the cosine annealed to that horizon, the twins reach 3.0955 and 3.0838—beating the recursive models by **0.0117 [0.0097, 0.0137] and 0.0088 [0.0067, 0.0111] nats while consuming two thirds of the compute**. Recursion wins only 128/488 and 160/488 windows here. At  $6\text{--}8\times$  the seed sd, the reversal is not noise. BLiMP agrees: against the annealed twins recursion is level ( $-0.02$  and  $-0.23$ ) and loses on the supplement ( $-4.65$  and  $-0.78$ ).

Figure 2 shows both axes together. The two axes separate cleanly. Per word of exposure, recursion wins: it reaches 3.1072 on 295M words where the twin needs 589M to reach 3.0955. Per FLOP, it loses. Which conclusion matters depends on which budget binds—and on a fixed-corpus benchmark, the data budget is the one that does.

### 5.4 What tying costs

Table 7 reports the result: untying is worth 0.0467 and 0.0399 nats—roughly twice the recursion gain itself, for three times the weights. Tying is thus a real compression: it forfeits about 0.04 nats and returns a  $3\times$  parameter saving. The clearest evidence that this trade is favourable here is across architectures rather than within a pair: **Recur-**

| Contrast            | $\Delta$ loss | 95% CI         | Won     |
|---------------------|---------------|----------------|---------|
| Untied vs Rec., 2:1 | +0.0467       | [.0447, .0488] | 485/488 |
| Untied vs Rec., 3:1 | +0.0399       | [.0379, .0419] | 485/488 |

Table 7: The price of weight tying. Positive favours the untied model, which holds  $3\times$  the learned parameters at identical depth, FLOPs, and initialization scale.

| Contrast               | $\Delta$ loss | 95% CI              | Won     |
|------------------------|---------------|---------------------|---------|
| 2:1 eff. vs uniq. init | +0.0034       | [.0024, .0044]      | 305/488 |
| 3:1 eff. vs uniq. init | +0.0003       | [ $-.0007$ , .0013] | 257/488 |

Table 8: Effective-depth versus unique-depth residual scaling, holding tying fixed. Positive favours effective-depth scaling.

**sive 2:1, at 46.9M non-embedding parameters, reaches 3.1072 against the 68.8M-parameter GQA baseline’s 3.1244**—better loss with 32% fewer learned weights (Figure 3). On BLiMP untying buys 2.61 points at 2:1 but nothing at 3:1 ( $+0.02$ ), while both untied arms gain 4–6 points on the supplement.

### 5.5 The initialization confound is small

In Table 8 the 3:1 contrast is a clean null: the interval spans zero and the sign test is at chance. The 2:1 contrast is small but detectable on evaluation-sample uncertainty—yet at roughly twice the seed sd of 0.0015, and with only 62.5% of windows won, it is not robustly identified either. BLiMP shows nothing in either family ( $-0.20$ ,  $-0.06$ ).

Effective-depth scaling therefore accounts for at most a seventh of the 2:1 recursion gain and none of the 3:1. The confound described in §3.2 exists but does not explain the result: **tying does the work**. We keep effective-depth scaling as the default on stability grounds rather than performance ones.

### 5.6 The crossover precedes the first epoch

Table 9 brackets the crossover. In the 2:1 pair it happens while both models are still inside their first pass over the corpus, so **repeated examples cannot be the mechanism there**. The 3:1 crossover sits near the boundary and remains ambiguous with respect to it. What varies systematically is effective depth: the 18-layer configuration crosses at  $\sim 100$ M tokens and the 24-layer one at  $\sim 175$ M, in both seeds. That ordering is what an early optimization cost predicts—deeper tied models pay more to get started—and is not what a repe-

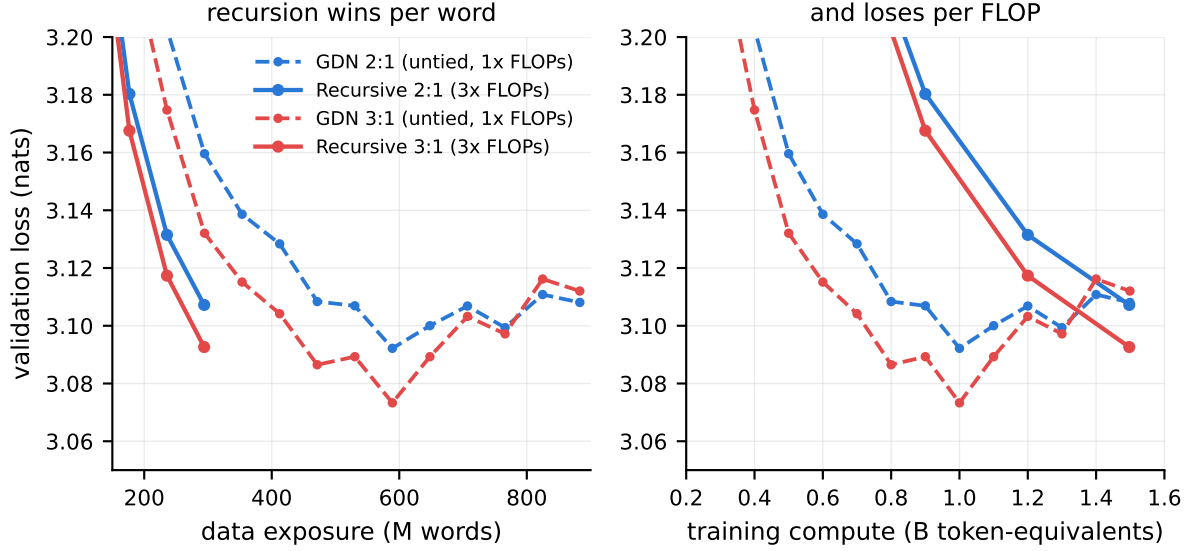


Figure 2: The same intervention against two budgets. **Left:** validation loss against data exposure. At equal exposure the recursive models (solid) sit below their untied twins (dashed); Recursive 2:1 reaches at 295M words what its twin needs  $\sim 590$ M words to match. **Right:** the same runs against training compute in token-equivalents, where a recursive model costs  $3\times$  per token. Here the twins’ curves run below the recursive endpoints, and both twins reach their minimum at  $\sim 1.0$ B token-equivalents before overtraining degrades them. Hue encodes the family (blue = 2:1, red = 3:1).

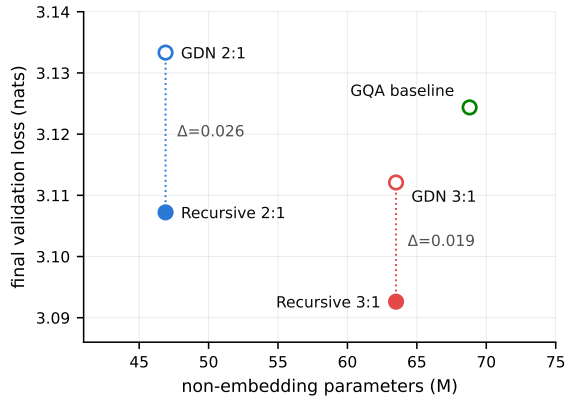


Figure 3: The parameter–quality trade-off for the primary grid. Vertical dotted lines connect the two parameter-matched pairs; filled markers are recursive models. Both recursive hybrids beat the larger pure-attention baseline on loss.

| Family                     | seed 43  | seed 44  | vs. ep. 1         |
|----------------------------|----------|----------|-------------------|
| 2:1 ( $D_{\text{eff}}$ 18) | 75–100M  | 100–125M | <b>before</b>     |
| 3:1 ( $D_{\text{eff}}$ 24) | 150–175M | 175–200M | <b>at / after</b> |

Table 9: Where the recursive model overtakes its twin, bracketed to 25M tokens. One corpus pass is 169.7M tokens.

| Pair | Suite  | s42   | s43   | s44   | mean         | sd   |
|------|--------|-------|-------|-------|--------------|------|
| 2:1  | BLiMP  | +3.31 | −1.38 | −0.18 | <b>+0.58</b> | 2.44 |
| 2:1  | Suppl. | +1.01 | −4.82 | +2.54 | <b>−0.42</b> | 3.88 |
| 3:1  | BLiMP  | +4.21 | +0.71 | +0.59 | <b>+1.84</b> | 2.05 |
| 3:1  | Suppl. | +1.24 | +4.20 | −0.25 | <b>+1.73</b> | 2.26 |

Table 10: Recursion advantage in macro-accuracy points, replicated over three seeds. Seed 42—the only seed evaluated in the original submission—is the most favourable of the three in both suites.

tition account predicts, since both models meet the epoch boundary at the same token count.

## 5.7 Grammatical accuracy does not replicate across seeds

Validation loss and BLiMP behave completely differently under seed replication, and the difference is large enough to change what this paper claims (Table 10).

The cause is not a defect of any one contrast

but the noise floor of the benchmark at this scale. Retraining a *single* architecture under a different seed, changing nothing else, moves BLiMP macro-accuracy by 1.70 to 5.56 points (Table 1, Appendix A). A 5.56-point swing from the seed alone is larger than any architectural difference in our grid, and it swallows two cross-architecture readings we previously drew. The pure-GQA baseline leads BLiMP at 71.07 while sitting tenth of thirteen on loss, a 3.59-point margin; and GDN 2:1

appeared to gain 5.32 points from exposure alone (62.24  $\rightarrow$  67.56, 295M to 884M words), which we had called the largest movement in the grid. Both are single-seed, and both lie inside the 5.56-point spread of one architecture, so we no longer claim either. For external scale the official BabyLM 2026 Strict GPT-2 reference reports 74.53 / 65.00, on a different architecture, recipe and exposure. The 2:1 recursion gain we previously reported at +3.31 is therefore withdrawn: it is one draw from a distribution whose spread exceeds it. The 3:1 gain keeps its sign in all three seeds and is the only accuracy claim we still make, at  $+1.84 \pm 2.05$ —directionally consistent, but with an interval that includes values near zero.

We state this as a finding rather than a limitation because the exposure is general: BabyLM-scale papers routinely compare architectures on single-seed BLiMP differences of one to three points, which our measurement says are not resolvable without replication.

**The other two official zero-shot suites are also null.** All thirteen 500M-token runs were also evaluated on the full official COMPS and entity-tracking sets. Recursion moves COMPS by  $-0.23$  (sd 0.45) and  $+0.21$  (sd 0.27), and entity tracking by  $-0.99$  (sd 1.41) and  $-0.10$  (sd 0.55); three of four contrasts change sign across seeds. More telling than the nulls is the absolute level: every model scores at or below the 20% chance level on entity tracking (16.79–19.11), far outside binomial noise over 6,780 items and not a length artifact—correct options and distractors both average 26.5 characters. The official GPT-2 reference reaches 23.58 there; COMPS sits at 53.5–54.7 against a 50% floor. Neither suite discriminates at this scale, so neither bears on the architectural question.

## 6 Discussion

### Recursion is a data-efficiency mechanism.

The two budget axes give opposite answers, and both are solid:  $+0.0252 / +0.0195$  nats per matched word,  $-0.0117 / -0.0088$  per matched FLOP. Tied re-application converts *additional computation* into quality at fixed parameter count, less efficiently than simply training the shallower model longer—but under a fixed corpus “longer” means repeating data, and repetition has a ceiling visible in our own runs. Recursion buys quality without spending exposure: where exposure is capped and compute is not, that is the useful trade;

outside it, it is not.

### Parameter efficiency survives on its own terms.

Recursive 2:1 beats a 47% larger pure-attention baseline on loss, and untying costs  $3\times$  the weights to buy 0.04 nats: under a weight-storage constraint—not a latency or FLOP one—recursion remains attractive, and §5.3 delimits rather than contradicts this. The 3:1 gap is the smaller of the two, most likely because its twin is less depth-starved at 8 unique layers and because the 24-deep tied model crosses over  $\sim 75$ M tokens later (§5.6), leaving less budget to amortize its early deficit.

**What the initialization result means.** We adopted effective-depth scaling on stability grounds and it delivered stability—21 runs, zero loss spikes, smooth residual-RMS growth across passes. It did not deliver much accuracy. Practitioners should treat depth-aware residual scaling as a stability measure, not a performance lever, and should not attribute reported recursion gains to it.

### Metric-dependent ranking, and which metric to trust.

Validation loss rewards depth and parameters; BLiMP rewards the all-attention baseline. BLiMP morphology shows the baseline’s largest margin (85.60 vs 78.80 for Recursive 3:1). We had read this divergence as an argument against selecting small models on dev loss alone. Seed replication inverts that advice: at this scale dev loss resolves architectures to  $\sim 0.001$  nats while BLiMP moves by several points on the seed alone (§5.7). The divergence is real, but the sharper reading is that one of the two metrics is measuring the architecture and the other is largely measuring the draw.

## 7 Conclusion

Across 21 controlled runs, weight-tied depth recursion improves parameter-matched validation loss by 0.0252 and 0.0195 nats—replicated over three seeds, positive in all six dev domains. Under matched compute the advantage reverses. Tying itself produces the effect: untying recovers 0.0467 and 0.0399 nats at  $3\times$  the parameters, and residual scaling accounts for at most 0.0034. No accuracy benchmark we ran resolves the contrast, so the claim rests on loss. Depth recursion pays where words are scarce and FLOPs are not—the regime BabyLM defines, and a narrower claim than “recursion helps.”

## Limitations

**Grammatical accuracy is unresolved, not merely unreplicated.** The paired models are now evaluated at three seeds (§5.7), and the result is that BLiMP’s seed noise at this scale (up to 5.56 points range for one architecture) exceeds the effects we set out to measure. We therefore make no 2:1 accuracy claim and only a weak 3:1 one. Every other model in Table 4—the untied, unique-init and compute-matched arms and the baseline—is still evaluated at seed 42 alone, so all cross-architecture BLiMP comparisons here inherit that floor and should be read as provisional.

**The checkpoint series is still on the narrow slice.** Final-checkpoint numbers are now domain-balanced over the full 17.4M-token dev set (§5.2), but the 255-checkpoint trajectories behind Figure 1, the crossover brackets of §5.6, and the compute-matched curves of §5.3 are still computed on the 2M-token prefix, to bound evaluation cost. Since the per-domain deltas are same-signed and similar in magnitude, we do not expect the trajectories to reorder, but we have not verified that at every checkpoint.

**Fine-tuning is not run.** We report the four official zero-shot suites (BLiMP, BLiMP Supplement, COMPS, entity tracking) but none of the fine-tuning tasks, which would require a Hugging Face wrapper around our native checkpoints and an `attention_mask` argument our GQA implementation does not accept—a change to the one shared primitive all 21 runs depend on.

**Compute matching is by token budget.** We match FLOPs analytically and verify against measured throughput (Appendix A), but did not tune the learning rate per budget; all runs share the  $6 \times 10^{-4}$  literature prior. A tuned longer run might favour the twins further.

**Narrow architectural slice.** One width (768), two sizes, two hybrid ratios,  $R = 3$  only. We do not vary  $R$ , and the untied arms exist at one depth each.

**Inference cost is untouched.** Naively unrolled decoding needs state for each application, increasing KV-cache and GDN recurrent-state storage. Parameter matching isolates quality per learned weight under a fixed weight-storage budget; it implies nothing about latency or inference-state memory.

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## A Reproducibility details

**Recipe.** Sequence length 4,096; batch 32 sequences = 131,072 tokens per optimizer step. AdamW (Loshchilov and Hutter, 2019) with  $\beta = (0.9, 0.95)$ , weight decay 0.1 on matrices only, gradient clipping 1.0; peak learning rate  $6 \times 10^{-4}$  with cosine decay to 10% of peak; bf16 autocast with fp32 master weights. The primary budget is 3,814 steps = 499,908,608 tokens  $\approx 2.9$  epochs  $\approx 295$ M words, with 250 warmup steps.

**Hardware.** Every run trains on a single H100-80GB (Modal); the full grid is  $\sim 12$  GPU-hours. All 21 runs completed with zero loss spikes (Figure 1).

**Code and artifacts.** Source code, model and variant definitions, the training and evaluation pipelines, and the evaluation artifacts behind every table and figure are available at <https://github.com/adityasasidhar/recursive-babylm>.

Table 11 reports per-run cost.

**Throughput confirms the FLOP match.** The untied deep arms are depth- and FLOP-matched to the recursive models by construction, and measured throughput agrees independently: 172,852 vs 180,660 tok/s at 2:1 and 133,451 vs 134,334 at 3:1. The compute-parity claim in §5.3 therefore does not rest on the analytic  $3\times$  alone.

**The evaluation-weighting pitfall, and its fix.** The original training-time validation logger split the 488 evaluation windows into batches of `micro_batch_size` and returned the mean of per-batch means. For micro-batch 16,  $488 = 30 \times 16 + 8$ , so the trailing 8 windows were weighted double; for micro-batch 8,  $488 = 61 \times 8$  exactly. The slice follows sorted-file order, and the last eight windows fall in an easier CHILDES region (1.60 nats against 3.15 for the preceding 480). Calibrating the five original runs against checkpoint evaluation recovers the bias exactly: +0.0246, +0.0248, +0.0245, +0.0245 nats for the four micro-batch-16 runs and  $-0.0000$  for the micro-batch-8 run. The logger now weights by rows, and the new runs’ logged values match checkpoint evaluation to four decimals. All paper numbers use uniform per-token weighting from checkpoints regardless of logger version. The lesson generalizes: *validation reductions must not depend on memory-layout settings*—and a correction calibrated on one code version must not be carried across a fix.

| Model                       | Tokens | Wall clock | Throughput (tok/s) | Peak mem (GiB) | Micro-batch | Spikes |
|-----------------------------|--------|------------|--------------------|----------------|-------------|--------|
| GQA baseline                | 500M   | 27.6 min   | 302,861            | 42.7           | 16          | 0      |
| GDN 2:1                     | 500M   | 18.8 min   | 443,084            | 33.6           | 16          | 0      |
| GDN 3:1                     | 500M   | 23.6 min   | 353,301            | 40.2           | 16          | 0      |
| Recursive 2:1               | 500M   | 46.2 min   | 180,660            | 70.0           | 16          | 0      |
| Recursive 3:1               | 500M   | 62.1 min   | 134,334            | 45.4           | 8           | 0      |
| Untied 2:1 deep             | 500M   | 48.3 min   | 172,852            | 37.1           | 8           | 0      |
| Untied 3:1 deep             | 500M   | 62.6 min   | 133,451            | 47.5           | 8           | 0      |
| Recursive 2:1 (uniqu. init) | 500M   | 46.5 min   | 179,427            | 70.0           | 16          | 0      |
| Recursive 3:1 (uniqu. init) | 500M   | 61.9 min   | 134,771            | 45.4           | 8           | 0      |
| GDN 2:1 (1B, annealed)      | 1B     | 36.1 min   | 462,782            | 33.6           | 16          | 0      |
| GDN 3:1 (1B, annealed)      | 1B     | 44.8 min   | 372,349            | 40.2           | 16          | 0      |
| GDN 2:1 (1.5B)              | 1.5B   | 53.3 min   | 469,173            | 33.6           | 16          | 0      |
| GDN 3:1 (1.5B)              | 1.5B   | 66.5 min   | 376,515            | 40.2           | 16          | 0      |

Table 11: Per-run cost. All runs:  $1\times$  NVIDIA H100 80GB HBM3 (Modal), CUDA 13.0, Python 3.11, flash-linear-attention 0.5.1, FlashAttention-3 for GQA layers, bf16 autocast, fp32 master weights, fused AdamW. Seed replicates (seeds 43, 44 of the four paired models) are omitted for space and match their seed-42 counterparts within measurement noise.

### Paired uncertainty.

paper/figures/paired\_uncertainty.py uses 100,000 paired nonparametric resamples with seed 20260714. Validation resamples the 488 aligned windows; BLiMP resamples aligned UIDs. Every contrast is signed so that positive favours the treatment arm. Bootstrap intervals measure only finite evaluation-sample uncertainty and assume exchangeable units despite domain/order correlations; the separate seed\_replication block reports across-seed spread, the only estimate of training variance here.

### BLiMP protocol.

src/common/blimp\_eval.py reproduces the official BabyLM 2026 causal scorer directly against native checkpoints: summed log-probability over every non-BOS sentence token, temperature 1, macro-averaged by UID (Warstadt et al., 2020). Evaluator commit 3d57ddc8 and data revision 8d52da94 are pinned and identical across all 13 evaluated models. Seven of 59,875 BLiMP pairs tied under every model and were resolved with a fixed seed.

**Extra logging for the recursive models.** Per-super-block gradient norms and per-recursion-pass activation RMS were logged throughout; activation RMS grows smoothly and roughly linearly across the three passes (Recursive 2:1:  $1.06 \rightarrow 1.60 \rightarrow 2.12$  in super-block 0 and  $2.79 \rightarrow 3.82 \rightarrow 5.01$  in super-block 1) with no sign of divergence, consistent with the  $1/\sqrt{2D_{\text{eff}}}$  residual initialization.